Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Thesis for the Master's degree in Physics

Abstract

[To be continued. . .]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Introduction to Hartree-Fock methods

In this chapter we give a technical introduction to the Hartree-Fock (HF) method and its numerical implementation. We will keep this discussion as general as possible, without referring directly to the EHM.

The HF method is a variational approach to approximate the ground-state wavefunction of a many-body hamiltonian. The wavefunction is determined by minimising energy, limiting our search to a specific subspace of states. The general idea is to approximate the real ground-state with the best possible many-body state that is an eigenstate of a non-interacting fermionic hamiltonian.

To start, we need to define what Hilbert subspace we work with. Then we need to identify a way of parametrising the possible non-interacting hamiltonians and their respective ground-states, and finally a method to choose the best approximation to our original model. This is done in Sec. 1.2, largely based on [29, 31].

Variational algorithms generally work by numeric minimisation of some given functional. This is not the case. We implement HF theory starting from a theoretical minimisation procedure, with some derivative being set to zero, and then prove that this procedure leads to solving a self-consistency problem. We end up with a set of equations where the right-hand side depends non-trivially on the left-hand side, and through an iterative approach we determine the solution. This procedure is sketched in detail in Sec. 1.3.

1.1 Prerequisites and generalities

In this short section we give some motivation to the choice of the HF method, before moving to a technical explanation of how the method works. As we said before, we are performing a constrained search for an approximation to the ground-state wavefunction.

Let $\hat{H}$ be a generic hamiltonian, and $\mathcal{H}$ the Hilbert space. For any many-body state $|\Psi\rangle \in \mathcal{H}$, the energy is given by the expectation value (EV)

$$E[\Psi] = \langle\Psi|\hat{H}|\Psi\rangle$$

Let $\mathcal{S} \subset \mathcal{H}$ be a subspace of states, ad let $|\Psi\rangle$ be a state inside that locally minimises energy,

$$|\Psi\rangle \in \mathcal{S} \qquad \frac{\delta E[\Psi]}{\delta\,|\Psi\rangle} = 0$$

If the true ground state does not belong to $\mathcal{S}$, the found local minimum $|\Psi\rangle$ approximates it. The energy $E[\Psi]$ is, in this case, bigger than the actual ground-state energy.

In HF methods, we identify $\mathcal{S}$ with the set of so-called "gaussian states". These are all the ground-states of non-interacting hamiltonians. We give a complete description of them in App. A. We restrict minimisation to gaussian states because we can use Wick's theorem to simplify significantly the computation of expectation values. Indeed, by averaging a quartic operator over a gaussian state, we get the exact decomposition

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle \overset{\mathrm{W}}{=} \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{1.1}$$

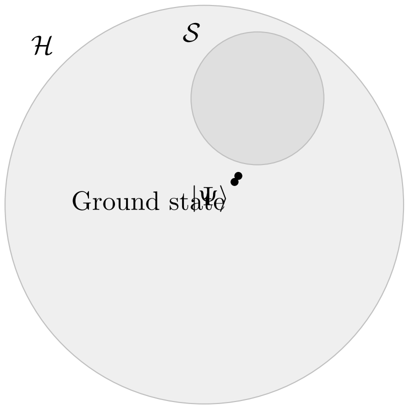

Figure 1.1

The subscripts indicate the names we will give to these terms throughout the text. The notation $\overset{\mathrm{W}}{=}$ indicates that we are using Wick's theorem.

Let us rephrase: the HF method is a minimisation constrained to all states that satisfy exactly Eq. (1.4). While this method simplifies a lot the computation of expectation values (reducing them to products of quadratic operators), the drawback is that the ground-state approximation is rather rough. Despite the roughness, HF methods are able to tackle various physical phenomena.

Operatively, we are approximating the hamiltonian with a non-interacting hamiltonian

$$\hat{H} \overset{\mathrm{HF}}{\simeq} \hat{H}_{\mathrm{HF}} = \sum_{\alpha} E_{\alpha} \hat{\gamma}_{\alpha}^{\dagger} \hat{\gamma}_{\alpha}$$

whose ground-state is obtained as a result of constrained minimisation. The notation $\overset{\mathrm{HF}}{\simeq}$ indicates that this operatorial approximation only holds inside the subspace of gaussian states. To determine the energies E_α as well as the fermionic operators $\hat{\gamma}_\alpha$ is the point of the HF method. These operators satisfy Fermi anti-commutation rules,

$$\left\{\hat{\gamma}_{\alpha}, \hat{\gamma}_{\beta}^{\dagger}\right\} = \delta_{\alpha\beta}$$

and are connected to the conventional electronic operators $\hat{c}_\alpha$ through an unitary map. This is discussed in detail in App. A.

1.1.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to antisymmetrize" a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r},\sigma)\psi_j(\mathbf{r},\sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume.

Let S_N be the $N!$ elements symmetric group of permutations of N objects and $\mathrm{P} \in \mathrm{S}_N$ a specific permutation of a given N-uple,

$$\mathrm{P} \quad : \quad (x_1, x_2, \cdots, x_N) \to \left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

Let F_{P} be the operator that implements the permutation P on the N variables of a function f,

$$F_{\mathrm{P}}\left[f(x_1, x_2, \cdots, x_N)\right] = f\left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

We define the antisymmetrization operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P}\in \mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} F_{\mathrm{P}}[f]$$

being $\sigma(\mathrm{P})$ the "parity factor" of the permutation $\mathrm{P} \in \mathcal{S}_N$. The factor $\sigma(\mathrm{P})$ counts the number of exchanges needed to obtain P: this way, an even number of exchanges gives a total sign $+1$, an odd number gives -1. By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots ; \mathbf{r}_N, \sigma_N) = \mathcal{A}\left[\prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i)\right]$$

This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots ; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \cdots, \psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \quad \rightarrow \quad \hat{c}_i, \hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N} \hat{c}_i^\dagger |\Omega\rangle \tag{1.2}$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

1.1.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (1.2) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (1.2) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \le i \le N \qquad & \hat{c}_i^\dagger |\Psi\rangle = 0 \\ i > N \qquad & \hat{c}_i |\Psi\rangle = 0 \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \le i \le N \\ \hat{c}_i & \text{for } i > N \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (1.2) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D} \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \qquad u_i^{(\ell)}, v_i^{(\ell)} \in \mathbb{C} \tag{1.3}$$

Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \cdots \hat{O}^{(2k)} \qquad k \in \mathbb{N}$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} + \cdots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j. Now, Wick's theorem states that

$$\begin{aligned} \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} = {} & \mathrm{N}\left\{ \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\} \\ & + \sum_{(\alpha\beta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} \mathrm{N} \left\{ \prod_{i \ne \alpha,\beta} \hat{C}_i \right\} \\ & + \sum_{(\alpha\beta)(\gamma\delta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} (-1)^{\zeta_{\gamma\delta}} D_{\gamma\delta} \mathrm{N} \left\{ \prod_{i \ne \alpha,\beta,\gamma,\delta} \hat{C}_i \right\} \\ & + \cdots \end{aligned}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha \hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \quad \to \quad \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle \Psi | \mathrm{N}\{\hat{A}_i \hat{A}_j \cdots\} | \Psi \rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle = \sum (\text{total sign factor}) \times (\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. **This is the defining property**

of gaussian states. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

1.1.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. **However, in order to treat some physical phenomena we need to extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators.** In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\boldsymbol{\gamma}} \\ \hat{\boldsymbol{\gamma}}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\gamma}}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If M is unitary, it follows:

$$\begin{aligned} \mathbb{I}_{2D\times 2D} &= MM^\dagger \\ &= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\ &= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ (U^\dagger)^2 + (V^\dagger)^2 & U^\dagger U + V^\dagger V \end{bmatrix} \end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}^\dagger_j \qquad \hat{\gamma}^\dagger_i \equiv \sum_{j=1}^{D} U^*_{ji}\hat{c}^\dagger_j + \sum_{j=1}^{D} V^*_{ji}\hat{c}_j$$

and substituting in the anti-commutator

$$
\begin{aligned}
\{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \{\hat{c}_{i'}^\dagger, \hat{c}_{j'}\} \\
&= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \delta_{i'j'} \\
&= \left[UU^\dagger + VV^\dagger\right]_{ij} \\
&= \delta_{ij}
\end{aligned}
$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij} \hat{\gamma}_j + \sum_{j=1}^{D} U_{ji}^* \hat{\gamma}_j^\dagger \qquad \hat{c}_i^\dagger \equiv \sum_{j=1}^{D} V_{ji}^* \hat{\gamma}_j^\dagger + \sum_{j=1}^{D} V_{ij} \hat{\gamma}_j$$

In Sec. 1.1.2 we argued that, when averaging a product of operators like the one of Eq. (1.3), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \to \hat{\gamma}_i, \hat{\gamma}_i^\dagger$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and "see" $|\Psi\rangle$ as a vacuum state.

As in Sec. 1.1.2, we conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{1.4}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov (HFB). For simplicity we will stick, from now on, to the name "HF method" referring to this discussion. **This argument has been carried out at zero temperature, but can be extended to finite temperature using appropriate path-integral techniques**.

1.2 Sketch of HF variational method

We have identified the features of the states we use to approximate the model ground-state. Now, let us get to the core of HF method and give a technical introduction on how it works. Consider the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \tag{1.5}$$

where

$$A_{\alpha\beta} = A^*_{\beta\alpha} \qquad B_{\alpha\beta\gamma\delta} = B^*_{\delta\gamma\beta\alpha} \tag{1.6}$$

to ensure hermiticity. Moreover, the model must show the following symmetry:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \tag{1.7}$$

because exchanging the position of the two annihilations and the two creations, the operator must remain the same.

Our aim is to approximate the above hamiltonian with something like:

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv E_0 + \sum_{\alpha,\beta} k_{\alpha\beta} \hat{\gamma}^\dagger_\alpha \hat{\gamma}_\beta \tag{1.8}$$

Old and new operators must be linked by an unitary map: with identical notation as in previous sections, $\hat{\Gamma} = M\hat{\Phi}$. To find the best approximation to the original hamiltonian means to identify:

- the offset E_0;
- the fermionic operators $\hat{\gamma}_\alpha$, $\hat{\gamma}^\dagger_\alpha$;
- the $k_{\alpha\beta}$ coefficients.

Note that, in terms of algebraic solution, E_0 is irrelevant. To determine it correctly is only relevant in order to get the correct approximation of the actual ground-state energy. Let us discuss briefly the features of $\hat{H}_{\mathrm{HF}}$ and then propose a solution.

1.2.1 Properties of the target quadratic hamiltonian

In this short section we enumerate some useful properties of $\hat{H}_{\mathrm{HF}}$. We will use also the following notation:

$$\hat{a}^-_\alpha \equiv \hat{c}_\alpha \qquad \hat{a}^+_\alpha \equiv \hat{c}^\dagger_\alpha$$

Due to linearity of the unitary mapping, the operator $\hat{H}_{\mathrm{HF}}$ of Eq. (1.8) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\mathrm{HF}} = E_0 + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta \tag{1.9}$$

By changing basis from $\hat{\gamma}$ to $\hat{c}$ operators, we moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ fields.

We can infer some mathematical properties of the fields $h^{pq}_{\alpha\beta}$. First, note that the hamiltonian can be represented through Nambu spinors as

$$\begin{aligned} \hat{H}_{\mathrm{HF}} &= E_0 + \hat{\Phi}^\dagger \mathbf{h} \hat{\Phi} \\ &= E_0 + \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix} \end{aligned}$$

Our hamiltonian must be hermitian. We conclude that:

$$\hat{H}^\dagger_{\mathrm{HF}} = \hat{H}_{\mathrm{HF}} \quad \Longrightarrow \quad h^{++} = \left[h^{--}\right]^\dagger$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^* \tag{1.10}$$

Same-sign off-diagonal fields are linked.

We can set, without loss of generality,

$$\mathrm{Tr}\, h^{+-} + \mathrm{Tr}\, h^{-+} = 0$$

This passage is guaranteed by the fact that if $\mathbf{h}$ is not traceless, we can redefine it by subtracting its trace and including the latter in the energy shift E_0. Then

$$\begin{aligned}
\hat{H}_{\text{HF}} &= E_0 + \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + h^{-+}_{\alpha\beta} \hat{c}_\alpha \hat{c}^\dagger_\beta \right] + (\text{same sign terms}) \\
&= E_0 + \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}_\beta \hat{c}^\dagger_\alpha \right) + h^{-+}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta \hat{c}_\alpha \right) \right] + (\text{same sign terms}) \\
&= E_0 - \sum_{\alpha,\beta} \left[h^{+-}_{\beta\alpha} \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha} \hat{c}^\dagger_\alpha \hat{c}_\beta \right] + (\text{same sign terms})
\end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta} \tag{1.11}$$

thus, also on-diagonal fields are linked. We are allowed to determine, say, just h^{+-}: its counterpart h^{-+} is completely determined. These properties were all we needed to get to the main argument of the method.

1.2.2 Self-consistent variational solution

We follow and extend the variational scheme proposed by Fabrizio [31]. Let:

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \right\rangle \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}_\beta \right\rangle \tag{1.12}$$

The first two are "normal densities"; the second two are "anomalous densities". Conjugation of the last two gives a symmetry relation similar to Eq. (1.10):

$$\left(\Delta^{++}_{\alpha\beta} \right)^\dagger = \Delta^{--}_{\beta\alpha} \tag{1.13}$$

The HF ground-state energy is given by:

$$\begin{aligned}
E_{\text{HF}} &= \left\langle \hat{H}_{\text{HF}} \right\rangle \\
&= E_0 + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta}
\end{aligned}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. Derive energy:

$$\begin{aligned}
\frac{\partial E_{\text{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta} \\
&= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}$$

Note that by construction E_{HF} is greater than the actual ground-state energy, since our approximated wavefunction is not the actual ground-state wavefunction. Now, from Hellman-Feynman theorem [29], we know that for a parametric hamiltonian it holds:

$$\begin{aligned}
\frac{\partial E_{\text{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle \Psi | \frac{\partial \hat{H}_{\text{HF}}}{\partial h^{rs}_{\mu\nu}} | \Psi \rangle \\
&= \langle \Psi | \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi \rangle \\
&= \left\langle \hat{a}^r_\mu \hat{a}^s_\nu \right\rangle \\
&= \Delta^{rs}_{\mu\nu}
\end{aligned}$$

The last two results imply:

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial\Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0 \tag{1.14}$$

This must hold for any possible choice of fields $h^{pq}_{\alpha\beta}$. Note that we did not impose the energy derivative to be zero: it is not the HF energy we are minimising.

Let us get back to the actual hamiltonian $\hat{H}$. To reduce it to a quadratic form is equivalent to assume that its ground-state has the properties described in previous sections. Then the following decomposition must hold:

$$\begin{aligned}
E &= \langle \hat{H}\rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta\rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{a}^+_\alpha \hat{a}^-_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\langle \hat{a}^+_\alpha \hat{a}^+_\beta\rangle\langle \hat{a}^-_\gamma \hat{a}^-_\delta\rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma\rangle\langle \hat{a}^+_\beta \hat{a}^-_\delta\rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta\rangle\langle \hat{a}^+_\beta \hat{a}^-_\gamma\rangle\right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma}\Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta}\Delta^{+-}_{\beta\gamma}\right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\alpha\beta}\Delta^{+-}_{\gamma\delta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta}
\end{aligned}$$

In second passage we inserted the the relations (1.12). Take the derivative:

$$\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\left(\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta}\frac{\partial\Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right) \\
+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\frac{\partial\Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta}\frac{\partial\Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right)
\end{aligned}$$

Using Eq. (1.7) we end up with:

$$\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right]\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\
+ \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2}\frac{\partial\Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial\Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}$$

Let now:

$$C_{\alpha\beta} \equiv \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right] \qquad D_{\alpha\beta} \equiv \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{1.15}$$

From Fermi anticommutation rules, we have:

$$\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \quad\Longrightarrow\quad \frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = -\frac{\partial\Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}$$

which implies:

$$\begin{aligned}
\sum_{\alpha,\beta} 2C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \\
&= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial\Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)
\end{aligned}$$

Moreover, using Eqns. (1.6), (1.13),

$$\sum_{\alpha,\beta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta} \left(\frac{B_{\delta\gamma\beta\alpha}\Delta^{--}_{\beta\alpha}}{2} \right)^*$$

Using these two relations, we conclude:

$$\sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} \frac{\partial\Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma} \frac{\partial\Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}$$

Recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \overset{!}{=} 0$$

The notation "$\overset{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} \left(C_{\alpha\beta} \frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha} \frac{\partial\Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta} \frac{\partial\Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^*_{\beta\alpha} \frac{\partial\Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \right) \overset{!}{=} 0 \tag{1.16}$$

Now compare Eqns. (1.14) and (1.16). We recognise $h^{+-}_{\alpha\beta} = C_{\alpha\beta}$ and $h^{++}_{\alpha\beta} = D_{\alpha\beta}$; the remaining fields are determined by the symmetry relations (1.11) and (1.10). Explicitly, the fields are self-consistently determined by the following two equations:

$$h^{+-}_{\alpha\beta} = \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \tag{1.17}$$

$$h^{++}_{\alpha\beta} = \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{1.18}$$

The right-hand side of both equations depends on the fields themselves: $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle \Psi[\mathbf{h}] | \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi[\mathbf{h}] \rangle \tag{1.19}$$

This set of equations defines our computational problem. We are searching a set of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ such that, determining the fields via Eqns. (1.17), (1.18), then Eq. (1.19) is respected. **The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.**

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not-so-short: by looking at Eq. (1.19), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (1.19).

1.2.3 Connection with mean-field theories

This short last section concludes the theoretical introduction to HF methods, and bridges the gap between HF approaches and mean-field theory (MFT). **In general, in MFT approaches we substitute the two-body (or more) interactions of the hamiltonian with a single-body interaction with a mean field, determined locally via self-consistency equations. This**

is, for example, the Weiß mean-field solution to the quantum Ising model. The HF method we sketched here does not work like that. In the context of Hubbard models, the analogous of Weiß MFT would be dynamical mean-field theory (DMFT) [10].

However, there are some points of connection. Let us sketch first the general scheme behind MFT. Suppose we have some operator $\hat{A} = \hat{B}\hat{C}$. Let

$$\delta\hat{O} \equiv \hat{O} - \langle\hat{O}\rangle \qquad \hat{O} \in \{\hat{A}, \hat{B}, \hat{C}\}$$

Then we have:

$$\begin{aligned}
\hat{A} &= \left(\langle\hat{B}\rangle + \delta\hat{B}\right)\left(\langle\hat{C}\rangle + \delta\hat{C}\right) \\
&= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\delta\hat{C} + \delta\hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\
&= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\left(\hat{C} - \langle\hat{C}\rangle\right) + \left(\hat{B} - \langle\hat{B}\rangle\right)\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\
&= -\langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}
\end{aligned}$$

Up to now, this relation is exact. MFT relies on the idea that the last term, of order $\mathcal{O}(\delta^2)$, can be neglected. This amounts to say that the operator fluctuations about its average are "small". This gives the approximation:

$$\hat{A} \simeq \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle - \langle\hat{B}\rangle\langle\hat{C}\rangle \tag{1.20}$$

Within the assumption that independently $\hat{B}$ and $\hat{C}$ fluctuations are "small", we are able to simplify $\hat{A}$ with a sum of three terms: the product of each operator with the averaged value of the other, minus the product of the averages.

In the context of HF methods, we are in an slightly different situation. We are not making assumptions on the fluctuations of the quartic sectors of the hamiltonian. We are constraining the space of wavefunctions and minimising energy there. Now, Eqns. (1.17), (1.18), although general and correct, are a little unpractical to use. To derive them was useful in order to prove that the self-consistent method is a variational search in disguise. We here sketch a more concrete method to obtain the approximated hamiltonian and connect it with MFT interpretation. What we derive here is a mere reformulation of the self-consistent result obtained above, presented in a somewhat lighter form.

Let us restart from the interacting hamiltonian of Eq. (1.5). In this thesis we deal with the EHM, thus (limitedly to this section) let us restrict to the two-body generic quartic interaction:

$$\hat{V} = \sum_{\alpha,\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha$$

with $\forall\alpha,\beta$: $V_{\alpha\beta} = V^*_{\beta\alpha}$. In terms of the general hamiltonian of Eq. (1.5), we have:

$$B_{\alpha\beta\gamma\delta} = 2V_{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{1.21}$$

We work with HF states. Thus Eq. (1.4) holds exactly (adapted for two indices instead of four)

$$\alpha \neq \beta \quad : \quad \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\rangle = \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle}_{\text{Hartree}} - \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\beta\hat{c}_\alpha\rangle}_{\text{Cooper}}$$

This gives:

$$\begin{aligned}
\langle\hat{V}\rangle &= \sum_{\alpha,\beta} V_{\alpha\beta}\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\rangle \\
&= \sum_{\alpha,\beta} V_{\alpha\beta}\left(\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\beta\hat{c}_\alpha\rangle\right)
\end{aligned} \tag{1.22}$$

from which we have the equations:

$$\frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle} = \delta_{\alpha\beta}\left[\sum_\gamma V_{\alpha\gamma}\langle\hat{c}^\dagger_\gamma\hat{c}_\gamma\rangle\right] - V_{\alpha\beta}\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle \tag{1.23}$$

$$\frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle} = V_{\alpha\beta}\langle\hat{c}_\beta\hat{c}_\alpha\rangle \tag{1.24}$$

Now, our aim is to approximate $\hat{V}$ as:

$$\hat{V} \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + y_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + y^*_{\beta\alpha}\hat{c}_\alpha \hat{c}_\beta \right] + (\text{some constant}) \tag{1.25}$$

where $x_{\alpha\beta} = x^*_{\beta\alpha}$ to ensure hermiticity. Note that we are actively ignoring terms like $\hat{c}\hat{c}^\dagger$. Anti-commuting these, we get sums of operators like $\hat{c}^\dagger\hat{c}$ and constants. The above expression is the most general. Take its EV:

$$\langle\hat{V}\rangle \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta}\langle\hat{c}^\dagger_\alpha \hat{c}_\beta\rangle + y_{\alpha\beta}\langle\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta\rangle + y^*_{\beta\alpha}\langle\hat{c}_\alpha \hat{c}_\beta\rangle \right] + (\text{some constant})$$

Now, taking the derivatives:

$$x_{\alpha\beta} = \frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle} \qquad y_{\alpha\beta} = \frac{\partial\langle\hat{V}\rangle}{\partial\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle}$$

Applying Eqns. (1.23) and (1.24), we get the identities:

$$x_{\alpha\beta} = \delta_{\alpha\beta}\left[\sum_\gamma V_{\alpha\gamma}\langle\hat{c}^\dagger_\gamma\hat{c}_\gamma\rangle\right] - V_{\alpha\beta}\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle \tag{1.26}$$

$$y_{\alpha\beta} = V_{\alpha\beta}\langle\hat{c}_\beta\hat{c}_\alpha\rangle \tag{1.27}$$

These equations are a rephrasing of Eqns. (1.17), (1.18) for the fields $h^{pq}_{\alpha\beta}$, setting $A_{\alpha\beta} = 0$. This can be checked by using Eq. (1.21).

Now: consider the special case where $\forall\alpha\colon V_{\alpha\alpha} = 0$. This is the case for the EHM. Then the above equations imply:

$$\begin{aligned}\hat{V} &= \sum_{\alpha\neq\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha \\ &\simeq \sum_{\alpha\neq\beta} V_{\alpha\beta}\Big(\overbrace{\hat{c}^\dagger_\alpha\hat{c}_\alpha\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\hat{c}^\dagger_\beta\hat{c}_\beta}^{\text{Hartree}}\overbrace{-\hat{c}^\dagger_\alpha\hat{c}_\beta\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\hat{c}^\dagger_\beta\hat{c}_\alpha}^{\text{Fock}} \\ &\qquad + \underbrace{\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\langle\hat{c}_\beta\hat{c}_\alpha\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\hat{c}_\beta\hat{c}_\alpha}_{\text{Cooper}}\Big) + (\text{some constant})\end{aligned}$$

To determine the constant, we take the EV of the above equation. The result should equal Eq. (1.22). Take e.g. the Hartree sector:

$$\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + C = \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle \quad \Longrightarrow \quad C = -\langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle$$

with C a constant. A similar implication holds for the Fock and the Cooper sectors. Then:

$$\begin{aligned}\hat{V} &= \sum_{\alpha\neq\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha & (1.28)\\ &\simeq \sum_{\alpha\neq\beta} V_{\alpha\beta}\Big(\hat{c}^\dagger_\alpha\hat{c}_\alpha\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\hat{c}^\dagger_\beta\hat{c}_\beta - \langle\hat{c}^\dagger_\alpha\hat{c}_\alpha\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\beta\rangle & (\text{Hartree})\\ &\qquad - \hat{c}^\dagger_\alpha\hat{c}_\beta\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\hat{c}^\dagger_\beta\hat{c}_\alpha + \langle\hat{c}^\dagger_\alpha\hat{c}_\beta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\alpha\rangle & (\text{Fock})\\ &\qquad + \hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\langle\hat{c}_\beta\hat{c}_\alpha\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\hat{c}_\beta\hat{c}_\alpha - \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\beta\hat{c}_\alpha\rangle\Big) & (\text{Cooper})\end{aligned}$$

Passing from the first row to the following ones, we decomposed the original quartic operator to a sum of three decomposition much similar to Eq. (1.20). This is the main connection with MFT. In HF theory, we do not write the quartic operator as a product of two quadratic operators and decompose it as we would do in MFT. Instead, a very similar decomposition (summed over the three sectors) is given by the fact that EVs are computed over HF states.

Let us summarise this section through a practical rule to decompose the quartic part of $\hat{H}_U$ and $\hat{H}_V$ in the EHM.

Local repulsion. Applying the HF method, the local repulsion $\hat{H}_U$ is approximated by the quadratic operator:

$$\begin{aligned}\hat{H}_U &= U \sum_i \hat{n}_{i\uparrow}\hat{n}_{i\downarrow} && (1.29)\\ &\simeq U \sum_i \Big(\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle + \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow} - \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle && \text{(Hartree)}\\ &\qquad - \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle - \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow} + \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle && \text{(Fock)}\\ &\qquad + \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\langle \hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle + \langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle \hat{c}_{i\downarrow}\hat{c}_{i\uparrow} - \langle \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\langle \hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle \Big) && \text{(Cooper)}\end{aligned}$$

Non-local attraction. Applying the HF method, the local repulsion $\hat{H}_V$ is approximated by the quadratic operator:

$$\begin{aligned}\hat{H}_V &= -V \sum_{\langle ij\rangle}\sum_{\sigma,\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} && (1.30)\\ &\simeq -V \sum_{\langle ij\rangle}\sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle + \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle && \text{(Hartree)}\\ &\qquad - \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} + \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\rangle\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\rangle && \text{(Fock)}\\ &\qquad + \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\langle \hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle + \langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle \hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\rangle\langle \hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle \Big) && \text{(Cooper)}\end{aligned}$$

We will use these decomposition referring to each sector in dedicated chapters.

1.3 Sketch of HF iterative algorithm

In this section we give a practical sketch of the algorithm we use to compute Eq. (1.19) for all given phases. We refer to this technique as "iterative Hartree Fock" (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i: then

$$\mathbf{v} \in \mathbb{C}^N \qquad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (1.19). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v})$$

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A}\colon \mathbb{C}^N \to \mathbb{C}^N \qquad \mathrm{A}(\mathbf{v}) = \mathbf{v} \qquad \Longrightarrow \qquad \mathbf{v} \text{ is a solution.} \tag{1.31}$$

We implement a rather easy scheme: first, we obtain an analytic expression for A. Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we "follow" the non-linear map A until we claim to have converged. Let us explain in detail:

1. Start with a generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta$$

   Choose the model parameters and set an electronic density;

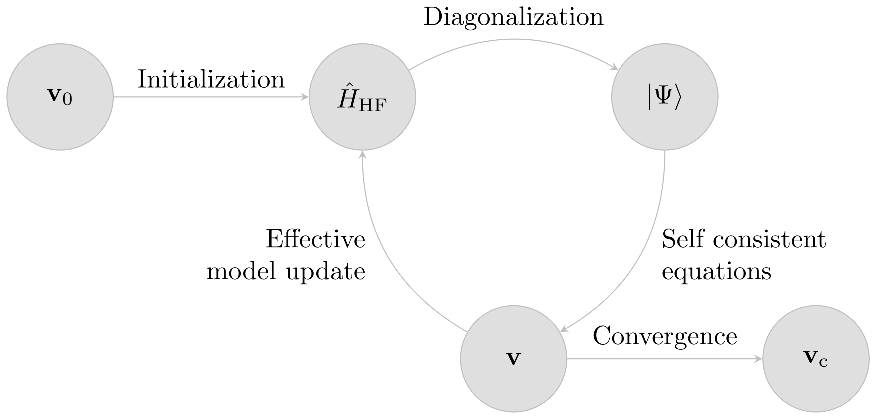

Figure 1.2 Schematisation of the self-consistent Hartree Fock method described in Sec. 1.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_{\mathrm{c}}$ the converged HFPs vector.

2. Following the discussion of Sec. 1.2, approximate analytically $\hat{H}$ with a quadratic hamiltonian

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha\beta} \left[h^{+-}_{\alpha\beta}\, \hat{c}^\dagger_\alpha \hat{c}_\beta - h^{+-}_{\beta\alpha}\, \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{++}_{\alpha\beta}\, \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \left(h^{++}_{\beta\alpha}\right)^* \hat{c}_\alpha \hat{c}_\beta \right]$$

with the fields given by Eqns. (1.17), (1.18);

3. Impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in. As an example: if we choose not to break total charge conservation, set $\forall\alpha, \beta\colon h^{++}_{\alpha\beta} = 0$;

4. Choose a starting value for the HFPs vector $\mathbf{v}_0$;

5. Set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 1.2:

   a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\mathrm{HF}}[\mathbf{v}]$;
   b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
   c) Diagonalise the effective hamiltonian $\hat{H}_{\mathrm{HF}}[\mathbf{v}] - \mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
   d) Compute $\mathrm{A}(\mathbf{v})$ using self-consistency equations;
   e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

We voluntarily left unspecified three details:

- How is the chemical potential μ determined?
- What criterion do we use to claim that the algorithm "converged"?
- How is the HFPs vector $\mathbf{v}$ updated?

The sections that follows deal with these questions.

1.3.1 Fixed density simulations and the role of μ

In this thesis all derivations and simulations are carried out at fixed density. This requires a way to fix μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v}, \mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v}, \mu) \equiv \frac{1}{\mathcal{V}} \sum_{\alpha} \langle \Psi[\mathbf{v}, \mu] | \hat{c}_\alpha^\dagger \hat{c}_\alpha | \Psi[\mathbf{v}, \mu] \rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0, 1]$. At step i, naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_{\mu} |n(\mathbf{v}_i, \mu) - n_0| \tag{1.32}$$

The density n must be a monotonically increasing function of μ: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu \,|_{\delta n(\mathbf{v}_i, \mu) = 0} \quad \text{being} \quad \delta n(\mathbf{v}_i, \mu) \equiv n(\mathbf{v}_i, \mu) - n_0$$

In order to where the function changes sign (which is, the position of μ_i) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;
2. Choose starting lower and upper bounds $\mu_{\mathrm{L}} < \mu_{\mathrm{U}}$;
3. Compute $\delta n(\mathbf{v}_i, \mu_{\mathrm{L}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i, \mu_{\mathrm{L}})| \leq \Delta n$, set $\mu_i = \mu_{\mathrm{L}}$ and halt;
   - if $\delta n(\mathbf{v}_i, \mu_{\mathrm{L}}) < -\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i, \mu_{\mathrm{L}}) > \Delta n$, shift the lower bound down, $\mu_{\mathrm{L}} \to \mu_{\mathrm{L}} - \Delta\mu$ and repeat from step 3;
4. Compute $\delta n(\mathbf{v}_i, \mu_{\mathrm{U}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i, \mu_{\mathrm{U}})| \leq \Delta n$, set $\mu_i = \mu_{\mathrm{U}}$ and halt;
   - if $\delta n(\mathbf{v}_i, \mu_{\mathrm{U}}) > \Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i, \mu_{\mathrm{U}}) < -\Delta n$, shift the upper bound up, $\mu_{\mathrm{U}} \to \mu_{\mathrm{U}} + \Delta\mu$ and repeat from step 4;
5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:
   a) Compute the midpoint,
   $$\mu_{\mathrm{M}} = \frac{\mu_{\mathrm{L}} + \mu_{\mathrm{U}}}{2}$$
   b) Evaluate:
      - If $|\delta n(\mathbf{v}_i, \mu_{\mathrm{M}})| \leq \Delta n$, set $\mu_i = \mu_{\mathrm{M}}$ and halt;
      - If $\delta n(\mathbf{v}_i, \mu_{\mathrm{M}}) > \Delta n$, set $\mu_{\mathrm{U}} = \mu_{\mathrm{M}}$ and repeat from point a);
      - If $\delta n(\mathbf{v}_i, \mu_{\mathrm{M}}) < -\Delta n$, set $\mu_{\mathrm{L}} = \mu_{\mathrm{M}}$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being F some function of the HFPs. The chemical potential μ is an example of Renormalised Model Parameter (RMP). As we will see, also the hopping t will turn out to be renormalised by interactions.

Note that at step i of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not μ_i. This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{H}|\Psi\rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimisation (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrised by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{K} + \mu N|\Psi\rangle\Bigg|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

In the first line μN is derived to zero; the constraint is imposed in the second line. Energy is given by:

$$\begin{aligned}\langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N\end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

1.3.2 The reject-mix-iterate convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of "convergence". Recall the definition of $\mathrm{A}(\mathbf{v})$, the non-linear map in HFPs space whose component are given by a set of self-consistency equations: we claim to have reached convergence when $\mathbf{v}$ comes "close enough" to a fixed point of the map A, as in Eq. (1.31). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \cdots)$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

being $\mathrm{A}(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and A_j its j-th component. The tolerance Δv_j is arbitrary. If just one component of $\mathbf{v}$ is mapped "too far" (with respect to Δv_j) the step is rejected and iterations continue. In this text we deal in with quantities of order 10^{-1}, so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step i, we check the distance of the two vectors $\mathrm{A}(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we need repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1 - g)\mathbf{v}_i \tag{1.33}$$

The "mixing parameter" g is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don't know how easy it is to actually get there following A. Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving "slower" ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling g. When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck in infinite loops. An example of this behaviour is given in Sec. **??**. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\cdots \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \cdots$$

With the notation "$\xrightarrow{\mathrm{A},g}$" we intend one algorithmic step at mixing parameter g. We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on g, where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found, however, was to just make g smaller. We gained empirical evidence that to "slow down" in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in "ordinary" situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound p to the total number of steps. If after p steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a "`NaN`" (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved and repeat simulations fine-tuning the parameters. This basically means to lower g and increase p. These considerations end this technical introductory chapter on iHF methods.

Appendix A

Gaussian states

A.1 Prerequisites and generalities

Let us start by discussing the properties of the many-body states we use to approximate the real ground-state of the system. We introduce Slater determinants to grasp the general idea behind HF variational method, and then extend to a larger set of states.

A.1.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to antisymmetrise" a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma \in \{\uparrow,\downarrow\}} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r},\sigma)\psi_j(\mathbf{r},\sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume.

Let S_N be the $N!$ elements symmetric group of permutations of N objects and $\mathrm{P} \in \mathrm{S}_N$ a specific permutation of a given N-uple,

$$\mathrm{P} \quad : \quad (x_1, x_2, \cdots, x_N) \to \left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

Let F_{P} be the operator that implements the permutation P on the N variables of a function f,

$$F_{\mathrm{P}}\left[f(x_1, x_2, \cdots, x_N)\right] = f\left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

We define the antisymmetrisation operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P} \in \mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} F_{\mathrm{P}}[f]$$

being $\sigma(\mathrm{P})$ the "parity factor" of the permutation $\mathrm{P} \in \mathcal{S}_N$. The factor $\sigma(\mathrm{P})$ counts the number of exchanges needed to obtain P: this way, an even number of exchanges gives a total sign $+1$, an odd number gives -1. By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \mathcal{A}\left[\prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i)\right]$$

This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1,\sigma_1;\mathbf{r}_2,\sigma_2;\cdots;\mathbf{r}_N,\sigma_N) = \frac{1}{\sqrt{N!}}\det\begin{bmatrix}\psi_1(\mathbf{r}_1,\sigma_1) & \psi_1(\mathbf{r}_2,\sigma_2) & \cdots & \psi_1(\mathbf{r}_N,\sigma_N)\\ \psi_2(\mathbf{r}_1,\sigma_1) & \psi_2(\mathbf{r}_2,\sigma_2) & \cdots & \psi_2(\mathbf{r}_N,\sigma_N)\\ \vdots & \vdots & & \vdots\\ \psi_N(\mathbf{r}_1,\sigma_1) & \psi_N(\mathbf{r}_2,\sigma_2) & \cdots & \psi_N(\mathbf{r}_N,\sigma_N)\end{bmatrix}$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1,\psi_2,\cdots,\psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \quad\rightarrow\quad \hat{c}_i,\hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N}\hat{c}_i^\dagger\,|\Omega\rangle \tag{A.1}$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

A.1.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (A.1) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (A.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0\le i\le N \qquad & \hat{c}_i^\dagger\,|\Psi\rangle = 0\\ i > N \qquad & \hat{c}_i\,|\Psi\rangle = 0\end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases}\hat{c}_i^\dagger & \text{for } 0\le i\le N\\ \hat{c}_i & \text{for } i>N\end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i,\hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (A.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D}\left(u_i^{(\ell)}\hat{c}_i + v_i^{(\ell)}\hat{c}_i^\dagger\right) \qquad u_i^{(\ell)},v_i^{(\ell)}\in\mathbb{C} \tag{A.2}$$

Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)}\hat{O}^{(2)}\cdots\hat{O}^{(2k)} \qquad k\in\mathbb{N}$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1\hat{C}_2\cdots\hat{C}_{2k} + \cdots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j. Now, Wick's theorem states that

$$\begin{aligned}\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k} = \mathrm{N}&\left\{\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\}\\ &+\sum_{(\alpha\beta)}(-1)^{\zeta_{\alpha\beta}}D_{\alpha\beta}\mathrm{N}\left\{\prod_{i\neq\alpha,\beta}\hat{C}_i\right\}\\ &+\sum_{(\alpha\beta)(\gamma\delta)}(-1)^{\zeta_{\alpha\beta}}D_{\alpha\beta}(-1)^{\zeta_{\gamma\delta}}D_{\gamma\delta}\mathrm{N}\left\{\prod_{i\neq\alpha,\beta,\gamma,\delta}\hat{C}_i\right\}\\ &+\cdots\end{aligned}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha\hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \quad\rightarrow\quad \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle\Psi|\mathrm{N}\{\hat{A}_i\hat{A}_j\cdots\}|\Psi\rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\rangle = \sum(\text{total sign factor}) \times (\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. **This is the defining property of gaussian states**. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle\hat{c}_\alpha^\dagger\hat{c}_\beta^\dagger\hat{c}_\gamma\hat{c}_\delta\rangle = \langle\hat{c}_\alpha^\dagger\hat{c}_\delta\rangle\langle\hat{c}_\beta^\dagger\hat{c}_\gamma\rangle - \langle\hat{c}_\alpha^\dagger\hat{c}_\beta\rangle\langle\hat{c}_\gamma^\dagger\hat{c}_\delta\rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

A.1.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. **However, in order to treat some physical phenomena we need to**

extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators. In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\gamma} \\ \hat{\gamma}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\gamma}$ and $\hat{\gamma}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If M is unitary, it follows:

$$\begin{aligned}
\mathbb{I}_{2D\times 2D} &= MM^\dagger \\
&= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\
&= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ \left(U^\dagger\right)^2 + \left(V^\dagger\right)^2 & U^\dagger U + V^\dagger V \end{bmatrix}
\end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}_j^\dagger \qquad \hat{\gamma}_i^\dagger \equiv \sum_{j=1}^{D} U_{ji}^*\hat{c}_j^\dagger + \sum_{j=1}^{D} V_{ji}^*\hat{c}_j$$

and substituting in the anti-commutator

$$\begin{aligned}
\{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \{\hat{c}_{i'}^\dagger, \hat{c}_{j'}\} \\
&= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \delta_{i'j'} \\
&= \left[UU^\dagger + VV^\dagger\right]_{ij} \\
&= \delta_{ij}
\end{aligned}$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{\gamma}_j + \sum_{j=1}^{D} U_{ji}^*\hat{\gamma}_j^\dagger \qquad \hat{c}_i^\dagger \equiv \sum_{j=1}^{D} V_{ji}^*\hat{\gamma}_j^\dagger + \sum_{j=1}^{D} V_{ij}\hat{\gamma}_j$$

In Sec. A.1.2 we argued that, when averaging a product of operators like the one of Eq. (A.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \to \hat{\gamma}_i, \hat{\gamma}_i^\dagger$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and "see" $|\Psi\rangle$ as a vacuum state.

As in Sec. A.1.2, we conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{A.3}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov (HFB). For simplicity we will stick, from now on, to the name "HF method" referring to this discussion. This argument has been carried out at zero temperature, but can be extended to finite temperature.

Appendix B

Connection with mean-field theories

This short last section concludes the theoretical introduction to HF methods, and bridges the gap between HF approaches and mean-field theory (MFT). **In general, in MFT approaches we substitute the two-body (or more) interactions of the hamiltonian with a single-body interaction with a mean field, determined locally via self-consistency equations. This is, for example, the Weiß mean-field solution to the quantum Ising model. The HF method we sketched here does not work like that. In the context of Hubbard models, the analogous of Weiß MFT would be dynamical mean-field theory (DMFT) [10].**

However, there are some points of connection. Let us sketch first the general scheme behind MFT. Suppose we have some operator $\hat{A} = \hat{B}\hat{C}$. Let

$$\delta\hat{O} \equiv \hat{O} - \langle\hat{O}\rangle \qquad \hat{O} \in \{\hat{A}, \hat{B}, \hat{C}\}$$

Then we have:

$$\begin{aligned}\hat{A} &= \left(\langle\hat{B}\rangle + \delta\hat{B}\right)\left(\langle\hat{C}\rangle + \delta\hat{C}\right) \\ &= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\delta\hat{C} + \delta\hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\ &= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\left(\hat{C} - \langle\hat{C}\rangle\right) + \left(\hat{B} - \langle\hat{B}\rangle\right)\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\ &= -\langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}\end{aligned}$$

Up to now, this relation is exact. MFT relies on the idea that the last term, of order $\mathcal{O}(\delta^2)$, can be neglected. This amounts to say that the operator fluctuations about its average are "small". This gives the approximation:

$$\hat{A} \simeq \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle - \langle\hat{B}\rangle\langle\hat{C}\rangle \tag{B.1}$$

Within the assumption that independently $\hat{B}$ and $\hat{C}$ fluctuations are "small", we are able to simplify $\hat{A}$ with a sum of three terms: the product of each operator with the averaged value of the other, minus the product of the averages.

In the context of HF methods, we are in an slightly different situation. We are not making assumptions on the fluctuations of the quartic sectors of the hamiltonian. We are constraining the space of wavefunctions and minimising energy there. Now, Eqns. (1.17), (1.18), although general and correct, are a little unpractical to use. To derive them was useful in order to prove that the self-consistent method is a variational search in disguise. We here sketch a more concrete method to obtain the approximated hamiltonian and connect it with MFT interpretation. What we derive here is a mere reformulation of the self-consistent result obtained above, presented in a somewhat lighter form.

Let us restart from the interacting hamiltonian of Eq. (1.5). In this thesis we deal with the EHM, thus (limitedly to this section) let us restrict to the two-body generic quartic interaction:

$$\hat{V} = \sum_{\alpha,\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha$$

with $\forall\alpha,\beta$: $V_{\alpha\beta} = V^*_{\beta\alpha}$. In terms of the general hamiltonian of Eq. (1.5), we have:

$$B_{\alpha\beta\gamma\delta} = 2V_{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{B.2}$$

We work with HF states. Thus Eq. (1.4) holds exactly (adapted for two indices instead of four)

$$\alpha \neq \beta \quad : \quad \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle}_{\text{Cooper}}$$

This gives:

$$\begin{aligned} \langle \hat{V} \rangle &= \sum_{\alpha,\beta} V_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \rangle \\ &= \sum_{\alpha,\beta} V_{\alpha\beta} \Big(\langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle \Big) \end{aligned} \tag{B.3}$$

from which we have the equations:

$$\frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle} = \delta_{\alpha\beta} \left[\sum_\gamma V_{\alpha\gamma} \langle \hat{c}^\dagger_\gamma \hat{c}_\gamma \rangle \right] - V_{\alpha\beta} \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle \tag{B.4}$$

$$\frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle} = V_{\alpha\beta} \langle \hat{c}_\beta \hat{c}_\alpha \rangle \tag{B.5}$$

Now, our aim is to approximate $\hat{V}$ as:

$$\hat{V} \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + y_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + y^*_{\beta\alpha} \hat{c}_\alpha \hat{c}_\beta \right] + (\text{some constant}) \tag{B.6}$$

where $x_{\alpha\beta} = x^*_{\beta\alpha}$ to ensure hermiticity. Note that we are actively ignoring terms like $\hat{c}\hat{c}^\dagger$. Anti-commuting these, we get sums of operators like $\hat{c}^\dagger\hat{c}$ and constants. The above expression is the most general. Take its EV:

$$\langle \hat{V} \rangle \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle + y_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle + y^*_{\beta\alpha} \langle \hat{c}_\alpha \hat{c}_\beta \rangle \right] + (\text{some constant})$$

Now, taking the derivatives:

$$x_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle} \qquad y_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle}$$

Applying Eqns. (1.23) and (1.24), we get the identities:

$$x_{\alpha\beta} = \delta_{\alpha\beta} \left[\sum_\gamma V_{\alpha\gamma} \langle \hat{c}^\dagger_\gamma \hat{c}_\gamma \rangle \right] - V_{\alpha\beta} \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle \tag{B.7}$$

$$y_{\alpha\beta} = V_{\alpha\beta} \langle \hat{c}_\beta \hat{c}_\alpha \rangle \tag{B.8}$$

These equations are a rephrasing of Eqns. (1.17), (1.18) for the fields $h^{pq}_{\alpha\beta}$, setting $A_{\alpha\beta} = 0$. This can be checked by using Eq. (1.21).

Now: consider the special case where $\forall \alpha$: $V_{\alpha\alpha} = 0$. This is the case for the EHM. Then the above equations imply:

$$\begin{aligned} \hat{V} &= \sum_{\alpha \neq \beta} V_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \\ &\simeq \sum_{\alpha \neq \beta} V_{\alpha\beta} \Big(\overbrace{\hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \hat{c}^\dagger_\beta \hat{c}_\beta}^{\text{Hartree}} \overbrace{- \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \hat{c}^\dagger_\beta \hat{c}_\alpha}^{\text{Fock}} \\ &\qquad + \underbrace{\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\beta \hat{c}_\alpha}_{\text{Cooper}} \Big) + (\text{some constant}) \end{aligned}$$

To determine the constant, we take the EV of the above equation. The result should equal Eq. (1.22). Take e.g. the Hartree sector:

$$\langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + C = \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle \qquad \Longrightarrow \qquad C = -\langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle$$

with C a constant. A similar implication holds for the Fock and the Cooper sectors. Then:

$$\begin{aligned} \hat{V} &= \sum_{\alpha \neq \beta} V_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha && \text{(B.9)} \\ &\simeq \sum_{\alpha \neq \beta} V_{\alpha\beta} \Big(\hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \hat{c}^\dagger_\beta \hat{c}_\beta - \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle && \text{(Hartree)} \\ &\qquad - \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \hat{c}^\dagger_\beta \hat{c}_\alpha + \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle && \text{(Fock)} \\ &\qquad + \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle \Big) && \text{(Cooper)} \end{aligned}$$

Passing from the first row to the following ones, we decomposed the original quartic operator to a sum of three decomposition much similar to Eq. (1.20). This is the main connection with MFT. In HF theory, we do not write the quartic operator as a product of two quadratic operators and decompose it as we would do in MFT. Instead, a very similar decomposition (summed over the three sectors) is given by the fact that EVs are computed over HF states.

Let us summarise this section through a practical rule to decompose the quartic part of $\hat{H}_U$ and $\hat{H}_V$ in the EHM.

Local repulsion. Applying the HF method, the local repulsion $\hat{H}_U$ is approximated by the quadratic operator:

$$\begin{aligned} \hat{H}_U &= U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} && \text{(B.10)} \\ &\simeq U \sum_i \Big(\hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle && \text{(Hartree)} \\ &\qquad - \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle && \text{(Fock)} \\ &\qquad + \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle \Big) && \text{(Cooper)} \end{aligned}$$

Non-local attraction. Applying the HF method, the local repulsion $\hat{H}_V$ is approximated by the quadratic operator:

$$\begin{aligned} \hat{H}_V &= -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} && \text{(B.11)} \\ &\simeq -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle && \text{(Hartree)} \\ &\qquad - \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle && \text{(Fock)} \\ &\qquad + \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle \Big) && \text{(Cooper)} \end{aligned}$$

We will use these decomposition referring to each sector in dedicated chapters.

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